

ASSIGNMENT 1

1. a. Find the least square line

t	2.0	3.0	5.0	6.0	9.0
s	7.0	17.0	49.0	71.0	161.0

$$y = mx + c$$

t	s	t^2	ts
2.0	7.0	4.0	14.0
3.0	17.0	9.0	51.0
5.0	49.0	25.0	245.0
6.0	71.0	36.0	426.0
9.0	161.0	81.0	1444.0
$\sum t = 25.0$	$\sum s = 305.0$	$\sum t^2 = 155.0$	$\sum ts = 2185.0$

$$n = 5$$

$$s = a_0 + a_1 t$$

$$\sum t(a_1) + ma_0 = \sum s$$

$$25(a_1) + 5a_0 = 305 \text{ -----(i)}$$

$$\sum t(a_0) + \sum t^2(a_1) = \sum ts$$

$$25(a_0) + 155(a_1) = 2185 \text{ ----(ii)}$$

$$25(a_1) + 5(a_0) = 305 \times 5$$

$$155(a_1) + 25(a_0) = 2185$$

$$125(a_1) + 25(a_0) = 1525$$

$$-155(a_1) + 25(a_0) = 2185$$

$$-30(a_1) = -660$$

$$a_1 = 22$$

From equation 1

$$25(a_1) + 5a_0 = 305$$

$$25(22) + 5a_0 = 305$$

$$550 + 5a_0 = 305$$

$$5a_0 = 305 - 550$$

$$5a_0 = -245$$

$$a_0 = -49$$

$$s = -49 + 22t$$

b. Polynomial Case

t	s	ts	t ²	t ² s	t ³	t ⁴
2.0	7.0	14.0	4.0	28	8	16
3.0	17.0	51.0	9.0	153	27	81
5.0	49.0	245.0	25.0	1225	125	625
6.0	71.0	426.0	36.0	2556	216	1296
9.0	161.0	1449.0	81.0	13041	729	6561
$\sum t = 25$	$\sum s = 305.0$	$\sum ts = 2185.0$	$\sum t^2 = 155.0$	$\sum t^2 = 17003$	$\sum t^3 = 1105$	$\sum t^4 = 8579$

$$5a_0 + 25a_1 + 155a_2 = 305$$

$$25a_0 + 155a_1 + 1105a_2 = 2185$$

$$155a_0 + 1105a_1 + 8579a_2 = 17003$$

$$\begin{array}{cccc} 5 & 25 & 155 & 305 \end{array}$$

$$\begin{array}{cccc} 25 & 155 & 1105 & 2185 \end{array}$$

$$\begin{array}{cccc} 155 & 1105 & 8579 & 17003 \end{array}$$

$$R_3 \longrightarrow R_3 - 31R_1$$

$$\begin{array}{cccc} 5 & 25 & 155 & 305 \\ 25 & 155 & 1105 & 2185 \\ 0 & 330 & 3774 & 7548 \end{array}$$

$$R_2 \longrightarrow R_2 - 5R_1$$

$$\begin{array}{cccc} 5 & 25 & 155 & 305 \\ 0 & 30 & 330 & 660 \\ 0 & 330 & 3774 & 7548 \end{array}$$

$$R_3 \longrightarrow R_3 - 11R_2$$

$$\begin{array}{cccc} 5 & 25 & 155 & 305 \\ 0 & 30 & 330 & 660 \\ 0 & 0 & 144 & 288 \end{array}$$

$$144a_2 = 288$$

$$a_2 = 2$$

$$0a_0 + 30a_1 + 330a_2 = 660$$

$$30a_1 + 330(2) = 660$$

$$30a_1 + 660 = 660$$

$$a_1 = 0$$

$$5a_0 + 25a_1 + 155a_2 = 305$$

$$5a_0 + 0 + 155(2) = 305$$

$$5a_0 = 305 - 310$$

$$5a_0 = -5$$

$$a_0 = -1$$

$$y = -1 + 0x + 2x^2$$

ASSIGNMENT 2

Write at least 10 applications of the least square line method

1. Scientific Research:

In scientific experiments, researchers often collect data where one variable depends on another. For instance, if you're testing how quickly a chemical reacts at different temperatures. The results are often scattered, but you want to know if there's a consistent trend like "does the reaction speed up as the temperature increases?"

By applying the least squares method, you can draw a line that represents the average behavior of the system. This helps to summarize complex data with a simple formula and can confirm or refute scientific theories.

For example: In physics, students often measure how far a spring stretches under different weights. Using the least squares line, they can find a clear relationship and determine the spring constant (Hooke's Law).

2. Economics and Finance:

Economists and financial analysts rely heavily on trends. Whether it's examining how consumer spending changes with income or predicting the growth of a country's GDP, they often work with large sets of data.

The least squares method allows them to create models that show how one factor influences another. These models are used to forecast the future, evaluate policies, or even detect market bubbles.

For example: A financial analyst could use it to predict future stock prices based on past performance, or see how interest rate changes affect home sales.

3. Health and Medicine:

In medical research and healthcare, it's common to study how lifestyle factors impact health. Things like how exercise might affect blood pressure, or how sleep duration relates to stress levels.

Using the least squares method, doctors and researchers can identify whether there's a meaningful link between two health indicators, and how strong that link is.

For example: A study might track the relationship between age and cholesterol levels across a population to better understand risk factors for heart disease.

4. Engineering:

Engineers deal with measurements and systems where precision is everything. But real-world data is rarely perfect, there's always some variation. That's where the least squares method comes in handy.

Whether while calibrating sensors, analyzing stress on materials, or designing electronic circuits, engineers use the least squares line to find the most accurate and reliable relationship between inputs and outputs.

For example: When measuring how pressure affects temperature in a gas system, the least squares line helps determine the system's behavior for safe and efficient operation.

5. Data Analytics and Machine Learning:

In the world of data science and machine learning, linear regression (which is based on the least squares method) is one of the most fundamental techniques.

Data analysts use it to train models that predict things like user behavior, sales growth, or customer churn. Even though more complex models exist today, linear models are still widely used because they are fast, interpretable, and surprisingly effective for many problems.

For example: Predicting housing prices based on square footage, number of rooms, and location using a regression model trained on historical data.

6. Marketing: Understanding What Drives Sales

Marketing teams constantly want to know what's working and what's not. For example, does spending more on online ads really bring in more customers? Or is product price the key factor driving sales?

By using the least squares line method, marketers can quantify how different variables (like ad budget, price, or social media engagement) affect sales, and make better decisions.

For example: A business could analyze past campaigns and discover that for every \$1,000 spent on social media ads, their sales increased by \$5,000 on average.

7. Transportation and Logistics:

In transportation, they deal with things like fuel efficiency, traffic flow, or delivery time all of which can be modeled with data. If there's a consistent pattern, the least squares line will help highlight it.

It's especially helpful in predicting and optimizing performance – such as how much fuel a truck uses for a certain route, or how traffic delays increase during peak hours.

For example: A logistics company might use the method to estimate how delivery time varies with distance and adjust route planning accordingly.

8. Environmental Studies:

Environmental scientists work with long-term data to study trends like climate change, air quality, and pollution levels. These variables naturally fluctuate, but the least squares method helps draw a clear line through the “noise” to reveal long-term changes.

This allows scientists and policymakers to take action based on real evidence of how the environment is changing.

For example: Using temperature records from the past 100 years, a least squares line can show the overall warming trend of the planet despite short-term ups and downs.

9. Education:

Educational researchers and school administrators are always trying to improve outcomes for students. They collect data on things like study time, class size, and teacher qualifications.

Using the least squares line, they can identify patterns like whether students who study more tend to perform better, or if a lower student-teacher ratio leads to higher scores.

For example: A school might use the method to show that students who complete weekly homework assignments score, on average, higher in tests.

10. Sports Analytics: Gaining the Winning Edge

In modern sports, data is everywhere from player stats to team performance. Coaches, scouts, and analysts use data to make decisions about training, recruitment, and strategy.

The least squares method helps them find relationships in performance data, such as how playing time affects goals scored, or how age affects performance decline.

For example: A basketball analyst might use it to examine how the number of assists relates to team wins across a season.

